ECE 555 Project — Milestone 1: Gate-Level Design

Single MLP neuron, ASAP7 (7 nm FinFET), Cadence Virtuoso · Fall 2026 · Team:

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1. What the neuron computes

The neuron computes **z = ReLU(y)**, where **$y = x_0 \cdot w_0 + x_1 \cdot w_1 + b$**. Inputs, weights and the bias weight are 2-bit two's-complement values restricted to {−1, 0, +1}, coded as `11`, `00` and `01`. The code `10` is never applied. That gives $-3 \le y \le +3$, so $z \in \{0, 1, 2, 3\}$ and the output is the 3-bit word Z2 Z1 Z0.

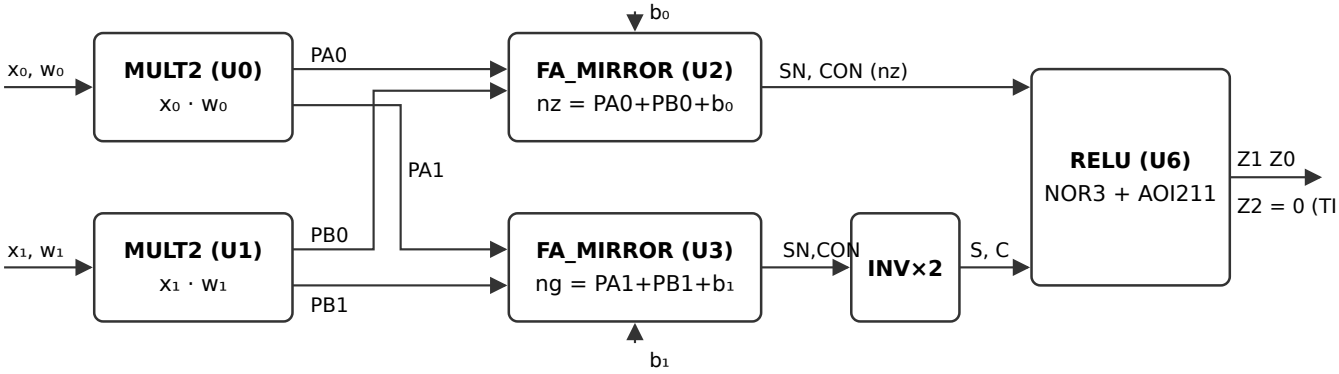
2. Architecture

Every term t (the two products and the bias) is itself in {−1, 0, +1}, and its two bits have a direct meaning. **$t_0 = 1$ means the term is nonzero**, and **$t_1 = 1$ means the term is negative**. If nz is the number of nonzero terms and ng is the number of negative terms, then

$$\textbf{y = (\#positive) − (\#negative) = nz − 2·ng}, \quad z = \max(0, nz - 2 \cdot ng).$$

So the datapath is (1) two **2-bit multipliers**, (2) one **full adder counting the three bit-0s** (nz) and one **full adder counting the three bit-1s** (ng), and (3) a small **ReLU logic block** that turns the two 2-bit counts into z.

This is the spec's multiplier → 2-bit adder → 3-bit adder → 3-bit 2:1 MUX chain in carry-save form. The two full adders add all three operands in one level with no carry ripple between them. The ReLU block merges the sign-selected MUX with the conversion from (nz, ng) to binary. The slides note that the example blocks are optional, and we chose this form because it has the shortest logic depth and the fewest transistors.



2.1 Truth tables

Multiplier truth table (legal codes)

x	x_1x_0	w	w_1w_0	p	p_1p_0
-1	11	-1	11	+1	01
-1	11	+0	00	+0	00
-1	11	+1	01	-1	11
+0	00	-1	11	+0	00
+0	00	+0	00	+0	00
+0	00	+1	01	+0	00
+1	01	-1	11	-1	11
+1	01	+0	00	+0	00
+1	01	+1	01	+1	01

$P0 = x_0 \cdot w_0 \quad P1 = x_0 \cdot w_0 \cdot (x_1 \oplus w_1)$

ReLU logic (all reachable counts, ng ≤ nz)

nz	c s	ng	c s	y	z	Z2Z1Z0
0	00	0	00	+0	0	000
1	01	0	00	+1	1	001
1	01	1	01	-1	0	000
2	10	0	00	+2	2	010
2	10	1	01	+0	0	000
2	10	2	10	-2	0	000
3	11	0	00	+3	3	011
3	11	1	01	+1	1	001
3	11	2	10	-1	0	000
3	11	3	11	-3	0	000

$Z1 = c_{nz} \cdot \bar{s}_{ng} \cdot \bar{c}_{ng} \quad Z0 = s_{nz} \cdot \bar{c}_{ng} \cdot (c_{nz} + \bar{s}_{ng}) \quad Z2 = 0$

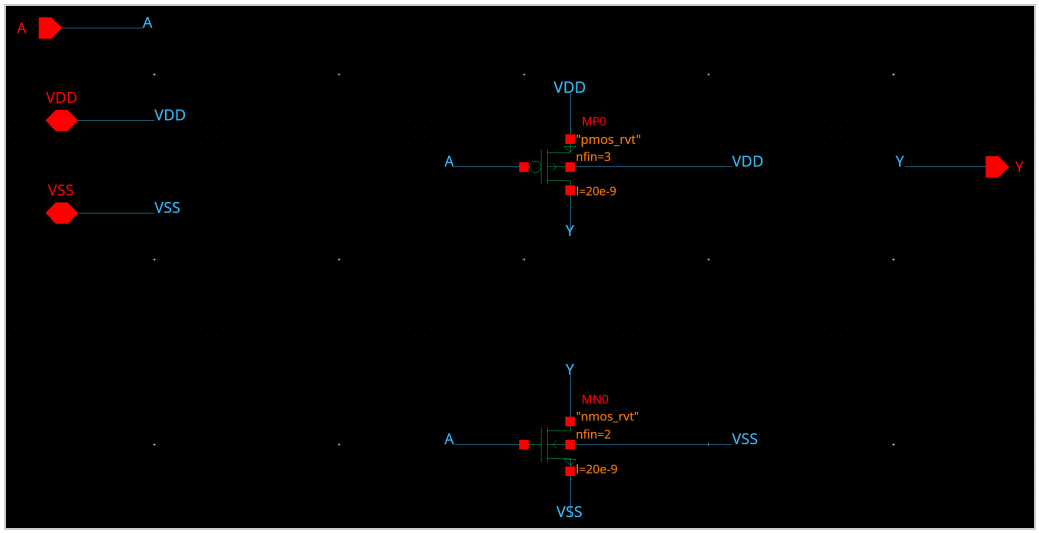
3. Sub-module schematics

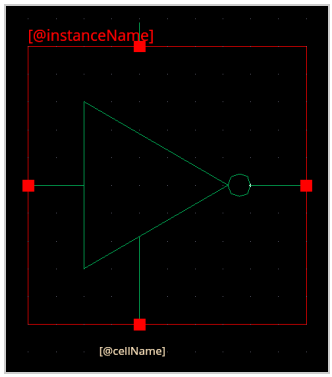
Each block is its own cell in library `NEURON`, with a schematic and a symbol, and higher levels are built only from those symbols. The leaf gates are static CMOS using ASAP7 RVT devices (`nmos_rvt` / `pmos_rvt`, L = 20 nm). VDD and VSS are explicit inputOutput pins rather than global nets, as Tutorial 2 recommends for LVS. The initial sizing is **2 fins per NMOS and 3 fins per PMOS**. Sizing will be refined in Milestones 2 and 3 against the (f_{max}/area) metric. Net names are shown as wire labels.

3.1 INV — Inverter

Function: Y = Ā · **2 transistors**

Used inside the multiplier (P0 = NOT n0) and to restore true-polarity sum and carry for the negative-count adder.



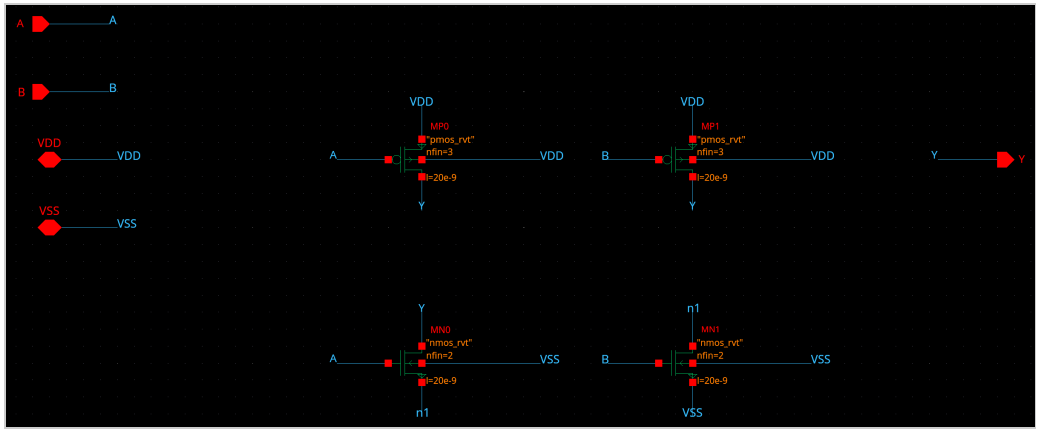


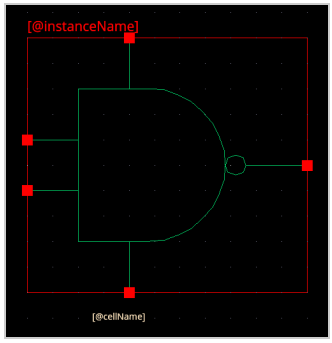
symbol

3.2 NAND2 — 2-input NAND

Function: Y = (A·B)̄ · **4 transistors**

Multiplier: n0 = (x₀·w₀)̄.



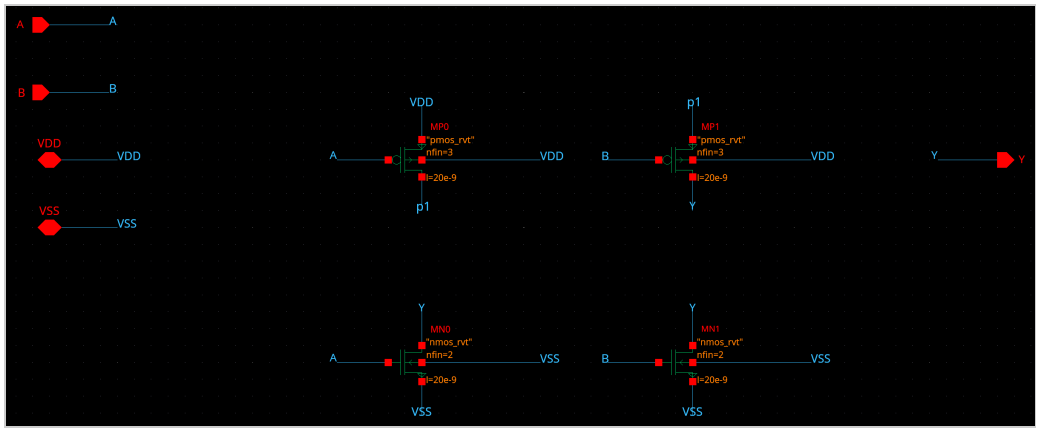


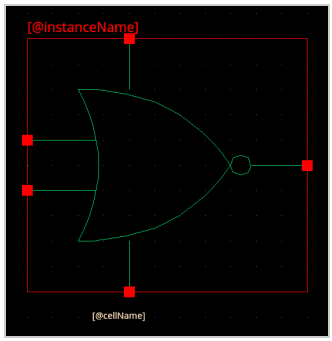
symbol

3.3 NOR2 — 2-input NOR

Function: $Y = \overline{(A + B)}$ · **4 transistors**

Multiplier: $P1 = \overline{(n0 + xn)} = x_0 w_0 (x_1 \oplus w_1)$.



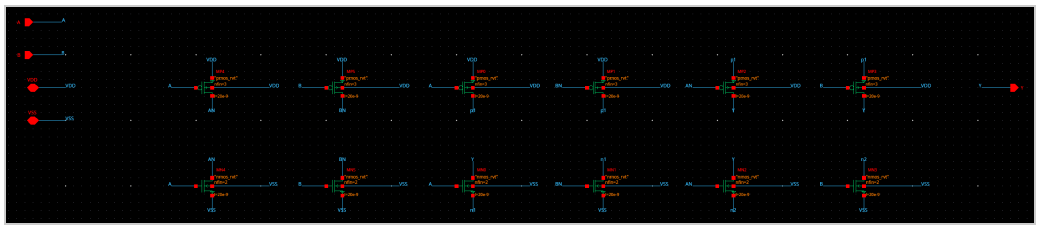


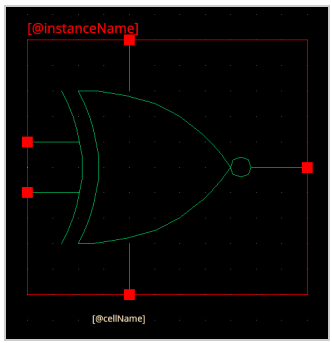
symbol

3.4 XNOR2 — 2-input XNOR (static, 12T)

Function: $Y = A \cdot B + \bar{A} \cdot \bar{B}$ · **12 transistors**

AOI22 core on $(A, \bar{B})$ and $(\bar{A}, B)$ with two input inverters. Multiplier sign: $xn = \overline{(x_1 \oplus w_1)}$.



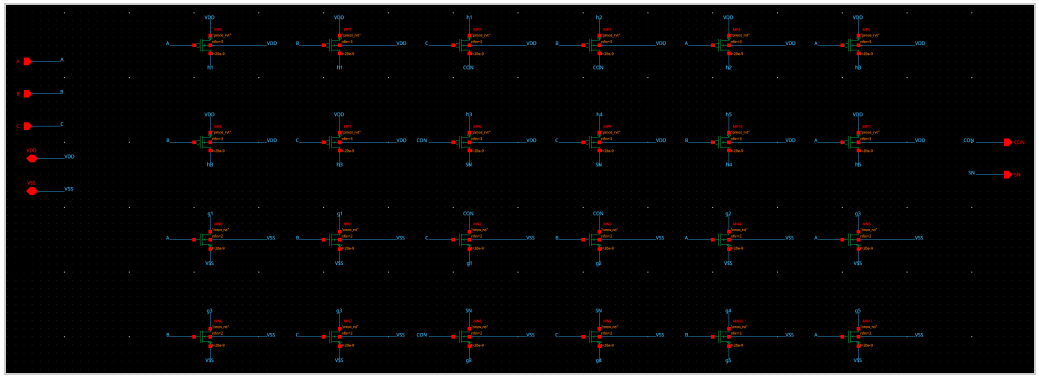


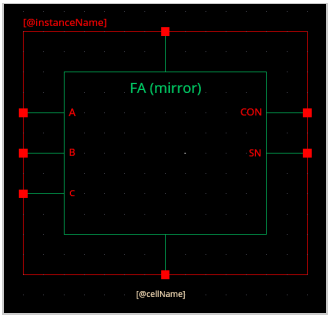
symbol

3.5 FA_MIRROR — 28T-style mirror full adder (24T core, inverted outputs)

Function: CON = (AB + C(A+B)̄), SN = (ABC + CON·(A+B+C)̄) · **24 transistors**

Counts how many of its three inputs are 1, giving the count in inverted form (CON = ¬carry, SN = ¬sum). C is the input closest to the output node, so the late-arriving product bit goes on C and the static bias bit goes on A.



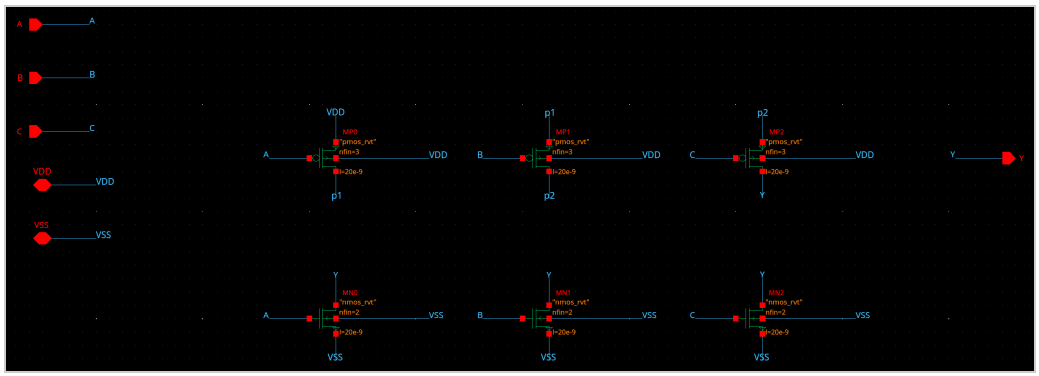


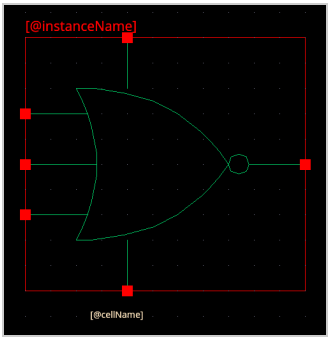
symbol

3.6 NOR3 — 3-input NOR

Function: Y = (A + B + C)̄ · **6 transistors**

ReLU logic: Z1 = (CON_{nz} + S_{ng} + C_{ng})̄.



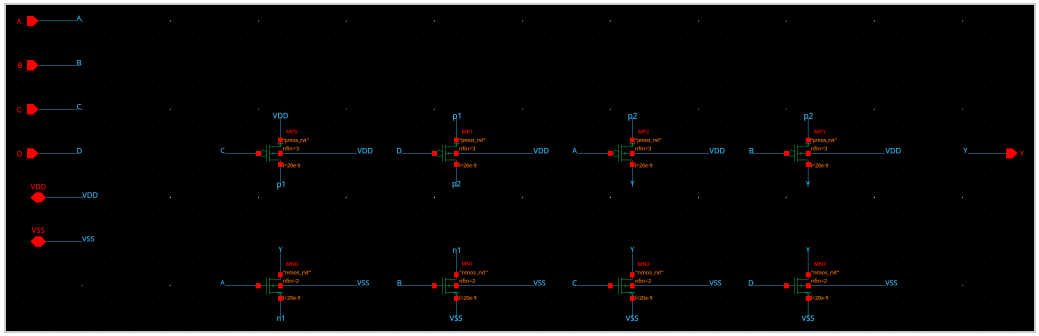


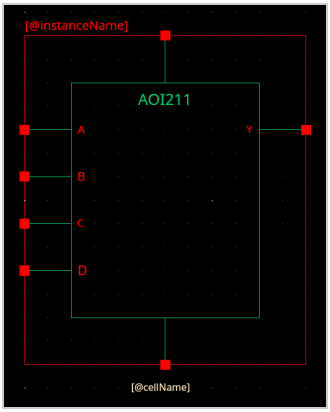
symbol

3.7 AOI211 — AND-OR-INVERT 2-1-1

Function: $Y = \overline{(A \cdot B + C + D)}$ · **8 transistors**

ReLU logic: $Z0 = \overline{(CON_{nz} \cdot S_{ng} + SN_{nz} + C_{ng})}$.



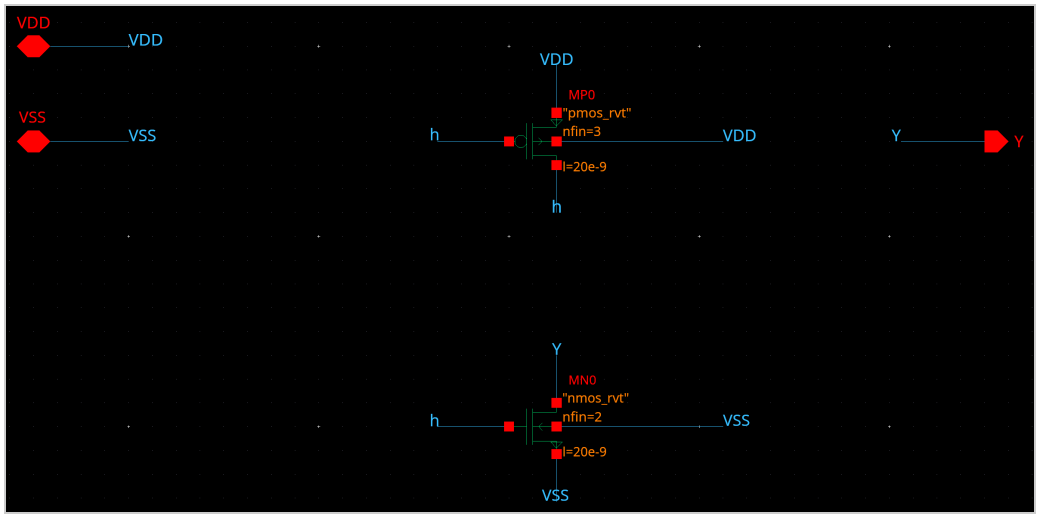


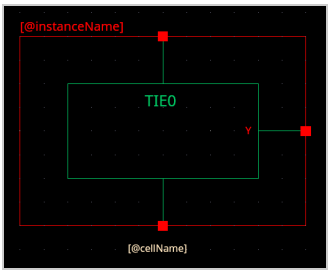
symbol

3.8 TIELO — Tie-low cell

Function: $Y = 0$ · **2 transistors**

Drives Z2 = 0 (the output can never exceed +3) without shorting the output pin straight to VSS.

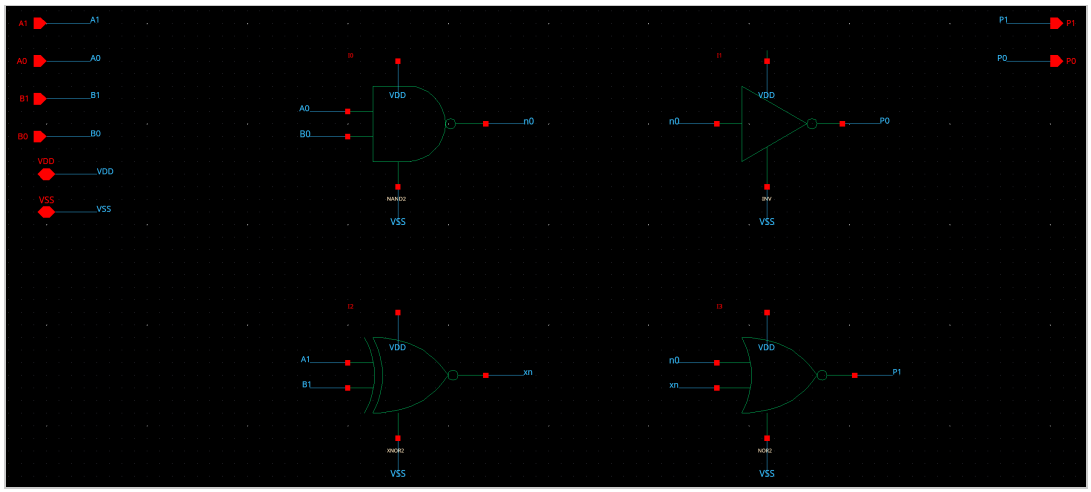


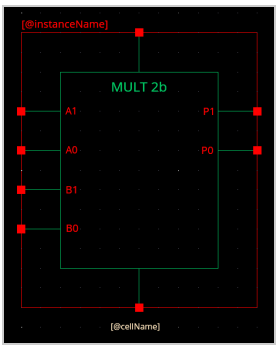


symbol

3.9 MULT2 — 2-bit × 2-bit multiplier (22 transistors)

The multiplier is NAND2 + INV for P0 and XNOR2 + NOR2 for P1. P1 reuses the NAND output n0, so the sign path is XNOR2 → NOR2, which is two gate levels.

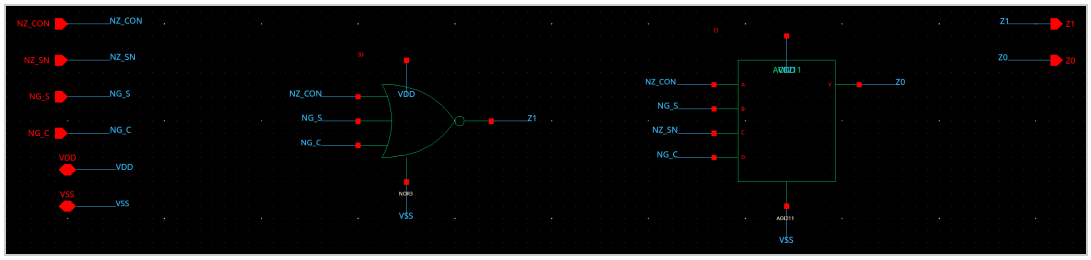


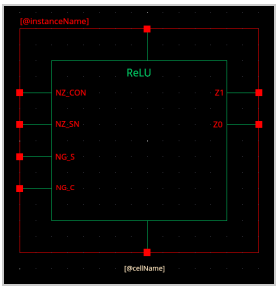


symbol

3.10 RELU — count-to-output ReLU logic (14 transistors)

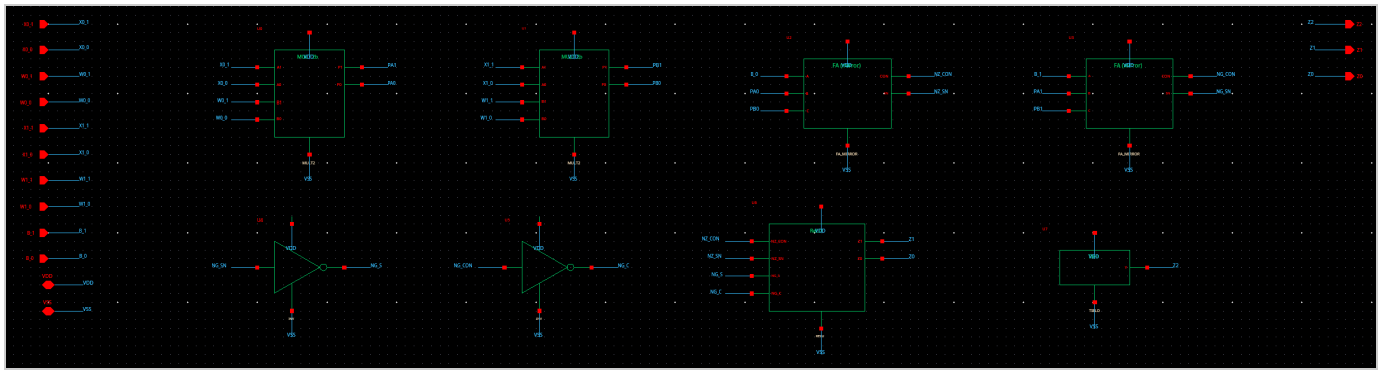
This block uses the inverted carry and sum of the nonzero counter directly, which saves two inverters, plus the true sum and carry of the negative counter.



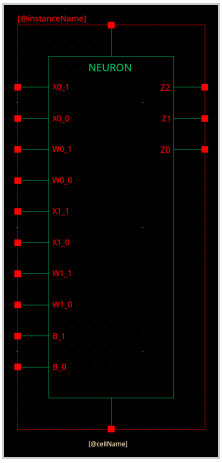


symbol

4. Top-level neuron



Block	Instances	Transistors each	Total
MULT2	2	22	44
FA_MIRROR	2	24	48
INV	2	2	4
RELU	1	14	14
TIELO	1	2	2
NEURON			**112**



NEURON symbol

Critical path (estimated): x_1/w_1 → XNOR2 (inverter + AOI22) → NOR2 → FA_MIRROR (C input) → INV → NOR3 / AOI211 → Z. The bias weight is static during operation, so it drives each full adder's slowest input (A), and the late product bits drive C.

5. Functional verification

MS1 does not require simulations, but we checked the design two ways before moving to layout. **Switch-level check:** the same netlists used to generate the Virtuoso cells were evaluated at switch level for every input of every leaf gate, and for all 243 legal (x_0, w_0, x_1, w_1, b) combinations of the full neuron. All 243 produce Z = max(0, y). We also confirmed that the connectivity read back from the saved Virtuoso schematics matches those netlists.

Pre-layout HSPICE check. The NEURON schematic was netlisted from Virtuoso (hspiceD) and simulated in HSPICE with the ASAP7 TT models at VDD = 0.9 V, 25 °C and 1 fF on each output. The testbench steps through all 243 legal input combinations (400 ps each) and samples Z2 Z1 Z0 at the end of each step. **243 of 243 cases match** max(0, y). Sampled output levels were at least 0.900 V for a logic 1 and at most 34 µV for a logic 0.